StepSpotter: OpenCV 5 decides what the agent does next

prepared by **Mikhail**

github.com/retailbox-automation/stepspotter-opencv · utgfmpcgxf.us-east-1.awsapprunner.com

> Every evaluation number in this report comes from `docs/EVAL.md` and `eval/results/*.json`, with the file and field next to it. Text in double braces is filled in at submission.

1. Problem

StepSpotter walks one person through one small home repair, one step at a time, on a photo of their own thing, and opens the next step only after a photo proves the last one was done. Until this entry, the proof was one vision-model call on one photo (`src/stepspotter/verifier.py:61`). A single-photo model check has three blind spots that a better prompt does not fix:

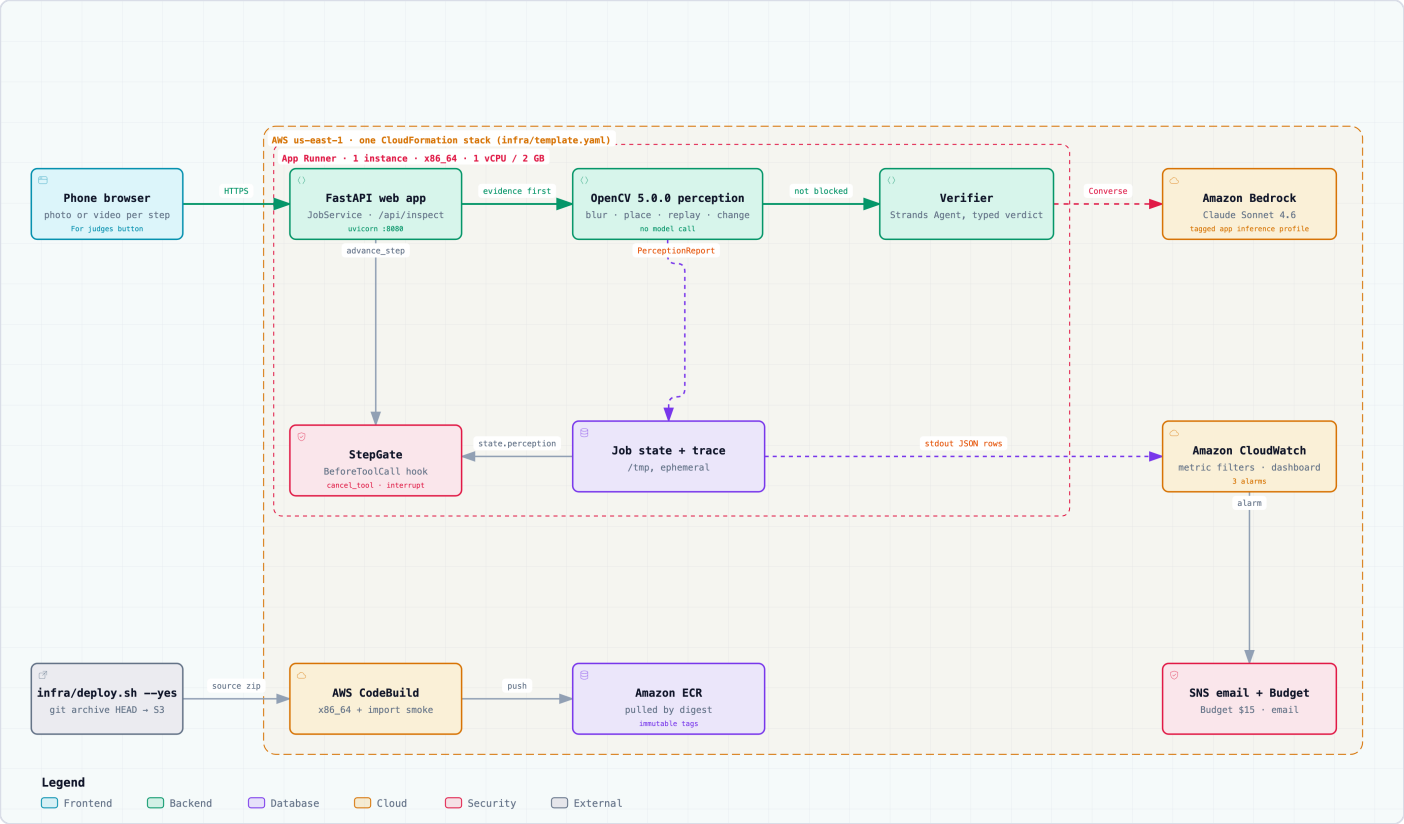
1. **A bad photo.** Blurred, dark, washed out, glare on the part, the part out of frame. The model either guesses or refuses without saying what to change, and every such attempt is a paid model call.
2. **The wrong photo.** One photo cannot tell the model that this is the same panel the job started on, or that this is the "before" photo sent again. A photo of a similar panel from somewhere else looks like valid proof.
3. **The wrong change.** The model does not compare before and after pixel by pixel: it cannot measure that the work zone changed and the do-not-touch zone did not.

OpenCV 5 now looks at every evidence photo first, without a model, and what it finds changes what the system does next: refuse with one instruction, send the photo to a one-photo or a two-photo model check, or stop the run for a person.

2. Users

People doing small, low-consequence jobs on their own home — low-voltage cabling, a network jack, a pressure-washer assembly — who learn best from their own photo marked up one step at a time. The first user is the author: the six evidence photos in this repository are of his own structured-wiring panel and wall jacks (`fixtures/perception/scenes.json`). StepSpotter's earlier build won the Everyday Agents Bronze award at the AWS "Agents for Humans" hackathon (see Disclosure); no other users are claimed. Jobs that need a licensed trade (mains wiring, gas, roofing, structural) are refused by the planner with an empty step list (`src/stepspotter/planner.py:26`, `:105`).

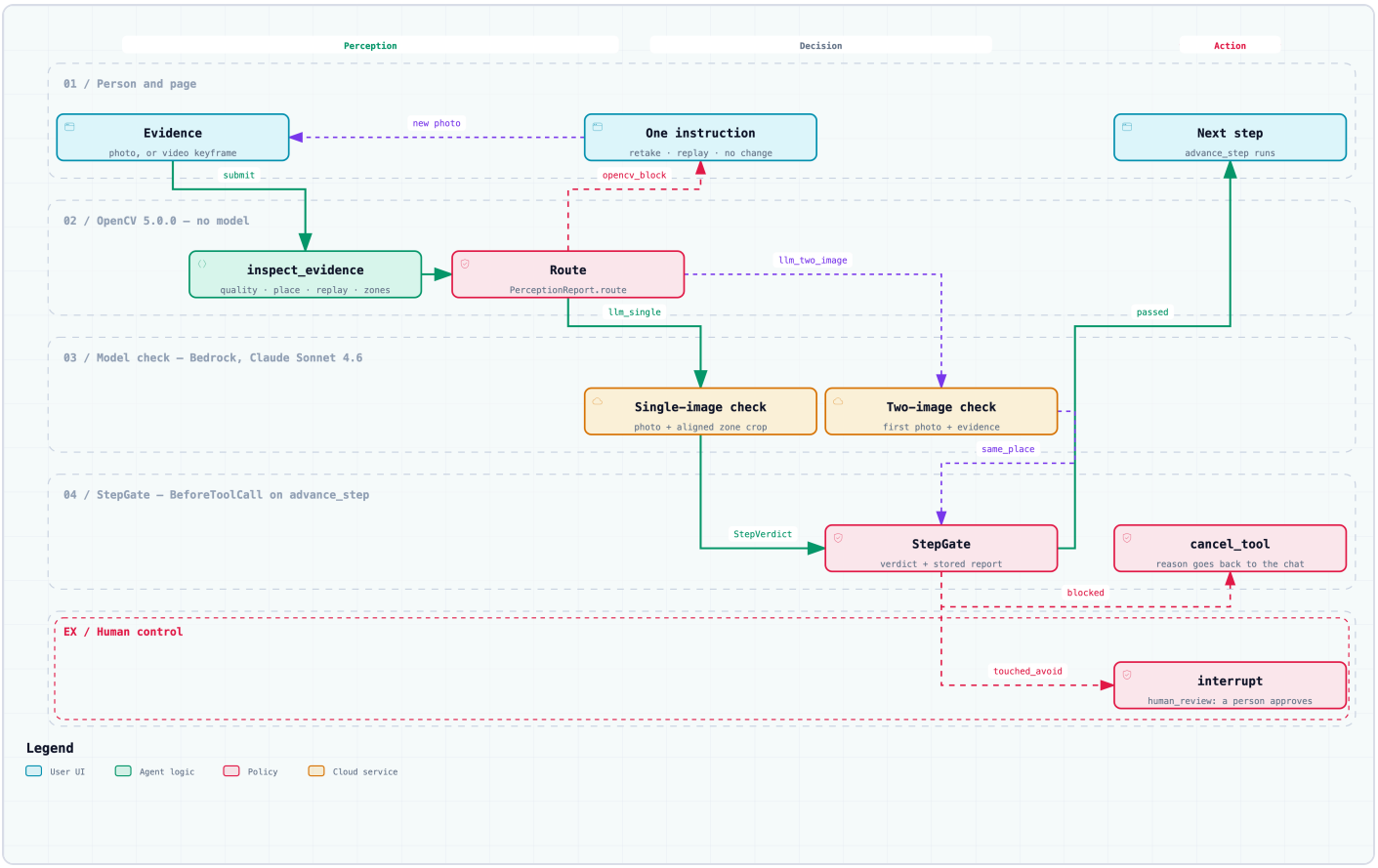
3. Architecture



Everything above runs in one container on AWS App Runner, deployed by one CloudFormation stack (`infra/template.yaml`). Every node in the diagram carries its source file and line in `docs/diagrams/architecture.html` (checked against commit `750d152`).

- **FastAPI web app** (`src/stepspotter/web/app.py`) serves the phone page and the API. A photo goes to `JobService.verify` (`src/stepspotter/guide.py:405`).
- **OpenCV 5.0.0 perception** (`src/stepspotter/perception/`) runs first on every evidence photo and returns a `PerceptionReport` with a decision and a route.
- **Verifier** — a Strands `Agent` on Amazon Bedrock (Claude Sonnet 4.6) that returns a typed `StepVerdict` (`src/stepspotter/vision.py:65`, `src/stepspotter/verifier.py:61`). It is called only for photos OpenCV did not refuse.
- **StepGate** — a Strands `BeforeToolCall` hook on the `advance_step` tool (`src/stepspotter/gate.py:64`). The page's "next step" button calls `advance_step` through a Strands `HookRegistry` with this hook (`src/stepspotter/web/gated.py:23`), the same dispatch the agent loop uses.
- **Job state and trace** live on the instance's disk (`/tmp/stepspotter`); perception rows are also printed to stdout as JSON for CloudWatch (`src/stepspotter/store.py:113`).

The agent loop, as the competition's Agentic Vision award asks for it — perception, decision, action:



Three front doors use the same tools and the same gate. On the default page the person's buttons drive the loop and code routes each photo. In **agent mode** (`/agent`, `src/stepspotter/web/agent_mode.py`) and in `stepspotter chat` (`src/stepspotter/cli.py:195`) the Strands Guide agent decides which tools to call — it can run `inspect_photo` before submitting (`src/stepspotter/guide.py:663`) — and every answer on the web shows the tool calls it made, what OpenCV decided about each photo, and any gate refusal. Agent mode is capped like the rest: starting a job counts as a job, a turn with a photo as a photo check, a text-only turn as an agent turn (60 a day for everyone, 20 an hour per address), and a hook cancels tool calls past 8 in one turn (`src/stepspotter/web/limits.py`, `src/stepspotter/web/agent_mode.py`).

4. OpenCV 5 implementation

Pinned: `opencv-python-headless==5.0.0.93`, `numpy==2.5.4` (`pyproject.toml`). The image build prints `cv2 5.0.0` from inside the built container (`infra/buildspec.yml:24`). Full description, every threshold and where it came from: `docs/PERCEPTION.md`.

check	OpenCV 5 calls	file
faces blurred before anything else reads the photo	`FaceDetectorYN` (YuNet, opencv_zoo), `GaussianBlur`	`perception/privacy.py:59`
sharpness, exposure, glare	`Laplacian`, `calcHist`, `threshold`, `connectedComponentsWithStats`	`perception/quality.py:65`, `:107`, `:80`, `:85`

check	OpenCV 5 calls	file
same place as the first photo	`SIFT_create` (on CLAHE luminance), `BFMatcher.knnMatch`, `findHomography(USAC_MAGSAC)`, `perspectiveTransform`, `isContourConvex`, `intersectConvexConvex`	`perception/imgio.py:88`, `scene.py:55`, `:61`, `:78`, `:94`, `:101`
the photo was already sent (replay)	homography near identity and aligned difference near zero	`perception/scene.py:247`
what changed in the work zone and in each do-not-touch zone	`warpPerspective`, `cvtColor(LAB)`, `createCLAHE`, `absdiff`, `morphologyEx`, `erode`, against the change outside all zones	`perception/change.py:40`, `:48`, `:54`, `:56`
a video instead of a photo	`VideoCapture`, best keyframe by sharpness × exposure × match	`perception/frames.py:30`, `:67`

The policy (`perception/inspect.py`, docstring, applied in order; the first rule that fires decides):

decision	route	next action
`retake` (blur, dark, washed out, glare, zone out of frame, blurrier than the first photo)	`opencv_block`	refused, one instruction to the person, **no model call**
`replay`	`opencv_block`	refused, no model call
`no_change` (the step had to change the work zone; it did not)	`opencv_block`	refused, no model call
`touched_avoid` (a do-not-touch zone changed)	`opencv_block`	the gate raises `interrupt("human_review")`
`place_unconfirmed` (OpenCV cannot line the photo up with the first photo)	`llm_two_image`	the model gets both photos and must answer `same_place`; the gate needs `same_place=true`
`ok`	`llm_single`	the model gets the photo and OpenCV's aligned crop of the work zone

Quality rules after the evaluation. The first evaluation run showed OpenCV refusing ordinary real photos of white things (basins, a white wall, a flashlight hot spot) as glare or overexposure. The rules were changed, calibrating on the calibration half of the evaluation set only (`src/stepspotter/perception/thresholds.py`):

- glare refuses a photo only when it **newly** covers at least 10 % of the work zone — saturated in the evidence but not at the same place in the first photo (`GLARE_ZONE_NEW_FRAC = 0.10`); anywhere else it is a note;
- overexposure is judged against the first photo of the place: the near-white share must exceed the first photo's by more than 0.12 (`OVEREXPOSED_MARGIN`) on top of the absolute 0.12 (`BRIGHT_FRAC_MAX`); with no readable first photo, only a photo more than half blown out is refused (`BRIGHT_FRAC_ABS_MAX = 0.50`);
- the relative-blur bar is 0.22 of the first photo's sharpness (`SHARP_RATIO_MIN`), and it is also read on a weak but sane fit of at least 12 matched points (`WEAK_FIT_INLIERS`), because blur is what removes matches.

On the calibration half, OpenCV alone then wrongly refused 1 of 103 good photos instead of 20 (`eval/results/opencv-before-glare-fix.json` and `eval/results/opencv.json`, `summary.by_split.calibration.false_blocks`).

Why `place_unconfirmed` is not a refusal. On our own photos, real re-shots of the same place from another angle kept 4–8 SIFT inliers after the homography fit, the same range as photos of different rooms (5–8); no inlier threshold separates them (`eval/results/opencv.md` at commit `ff97511`, run on 2026-10-21 under the previous policy, which refused 3 of 5 such re-shots). So OpenCV confirms a place when it can (≥ 25 inliers, `perception/thresholds.py:152`) and otherwise hands the question to the model with both photos — a dearer check, never a refused person. The gate fails closed: a pass without `same_place=true` cannot close the step (`guide.py:449`, `gate.py:153`).

Where OpenCV output changes a later decision — the evidence the Agentic Vision award asks for, each with a test:

1. A blocking decision becomes a failed verdict with `source="opencv"` and the model is never called (`guide.py:430`; `tests/test_perception_gate.py`, `test_a_blurred_photo_is_refused_without_calling_the_model`).
2. The route chooses which model call is made: one photo plus crop, or two photos with a required `same_place` (`guide.py:443`; `test_a_photo_opencv_cannot_line_up_goes_to_the_two_photo_check`).
3. The gate cancels `advance_step` on a stored blocking report even beside a passing verdict (`gate.py:141`; `test_the_gate_refuses_on_a_blocking_report_even_beside_a_passing_verdict`).
4. A change in a do-not-touch zone turns the next `advance_step` into a request for a person's approval (`gate.py:91`; `test_a_change_in_the_do_not_touch_zone_stops_the_run_for_a_person`).
5. `inspect_photo` lets the agent ask once for a photo from the first angle before it submits (`guide.py:663`; `test_inspect_photo_offers_one_ask_for_the_first_angle_and_never_refuses`).
6. After three OpenCV blocks in a row on one step, the next `retake` or `no_change` goes to the model with OpenCV's findings attached, so a false positive cannot trap a person (`perception/inspect.py`, `apply_stuck_rule`; `tests/test_perception_failures.py`).

A recorded live agent run (`data/demo/agentic-opencv/`: `transcript.md`, `trace.jsonl`, `NOTES.md`). The real Guide agent (Strands, Claude Sonnet 4.6 on Bedrock) with a scripted person, on 2026-10-21:

1. a blurred copy of the panel → the agent called `inspect_photo`; OpenCV said `retake` (sharpness 4, needs 45) in 44 ms with no model call; the agent relayed "hold the phone still" and did not submit;
2. a photo of a wall jack, not the panel → OpenCV could not line it up (`place_unconfirmed`); the person insisted, the agent submitted, and the two-photo check (`llm_two_image`) answered `same_place=false` — not passed;
3. the panel from across the room → OpenCV confirmed the place (52 matched points), the single-photo check (`llm_single`) passed it, the agent called `advance_step`, the gate allowed it, and step 2 was shown;
4. on the permissive prompt ("the user is always right"), the agent's `advance_step` on step 2 was cancelled by the gate hook in code, because step 2 had no passed photo. This was recorded as a follow-up turn on the same saved job, after the first attempt's permissive agent asked for the step number instead of trying to skip.

The plan was fixed from the repository fixture (`plan.json`) so that archive photos could honestly pass a step; the agent, the Marker, the verifier, OpenCV and the gate were live. 19 Bedrock calls in all (`run.json` → `total_with_follow_up.bedrock_calls`). The first recording, where no step could pass, is kept unchanged in `run-1/`.

Every OpenCV decision is one trace row with `decision`, `route`, `llm_called` and `ms_total` (`guide.py:320`). On the page, "Everything it did" shows it as "OpenCV photo check (code, not the model)" with the match lines, the change heatmap and the time.

{{SCREENSHOT:trace — docs/screens/trace-opencv.png, a judge scenario's trace row on the live URL}}

5. AWS deployment

component	what it does here	where
AWS App Runner	one instance, 1 vCPU / 2 GB, x86_64, min = max = 1 (job state is on local disk)	`infra/template.yaml:440`, `:450`
Amazon Bedrock	Claude Sonnet 4.6 through a tagged application inference profile, so model spend is attributable to the stack	`infra/template.yaml:355`
IAM	instance role may invoke one model and put metrics into one namespace; no secrets in the environment	`infra/template.yaml:393`
CloudWatch	four metric filters on the JSON trace (`PerceptionDecisions` by decision, `LlmCallsSaved`, `LlmCallsMade`, `PerceptionLatencyMs`), three alarms (5xx, request flood, perception p95), one dashboard	`infra/template.yaml:502` – `654`
SNS + AWS Budget	alarm email; $15/month Budget created before the service	`infra/template.yaml:160`, `:177`
CodeBuild + ECR + S3	`git archive HEAD` → S3 → x86_64 build with an architecture check and an import smoke → ECR (immutable tags, scan on push) → App Runner by digest	`infra/deploy.sh:156`, `infra/buildspec.yml`, `infra/template.yaml:212`, `:322`

One command deploys (`infra/deploy.sh --yes`) and one removes (`infra/destroy.sh --yes`); full procedure, measured durations and cost in `docs/DEPLOY.md`, operations and the kill switch in `docs/OPERATIONS.md`. Spend is bounded in code: per-address caps (6 jobs, 30 photo checks, 20 agent turns an hour) and global daily caps (6 jobs, 30 photo checks, 60 agent turns a day); jobs and photo checks alone bound the default page at ≤ 78 Bedrock calls a day (`src/stepspotter/web/limits.py`, `docs/OPERATIONS.md`). A photo OpenCV refuses costs no model call. Model price used: $3 / $15 per million input / output tokens for Claude Sonnet 4.6 in us-east-1 (AWS Price List API, effective 2026-09-01, read 2026-10-21); at the 3,142 / 657 average tokens per call measured on the first deployment that is about $0.019 per model call (`docs/DEPLOY.md`).

Deployed build: {{DEPLOY:commit and image digest of the final redeploy}}. The timings in §6 were measured on a shared Mac, not on the App Runner instance (1 vCPU); the evaluation did not measure OpenCV latency on AWS.

6. Evaluation

Full method, data provenance, every failure with pictures and the command behind every number: `docs/EVAL.md`. Result files: `eval/results/*.json`.

Data. 548 cases (`eval/cases/manifest.json`): 90 real and 458 synthetic, from 6 photos of our own house, 40 Wikimedia Commons photos with API-checked licences (`eval/cases/public/ATTRIBUTION.md`) and 16 frames of the author's own under-sink video. Split by scene or object into a calibration half (308) and a test half (240); no threshold was tuned on the test half.

Configurations. A = the model on the evidence photo alone (the gate before OpenCV) · A′ = the model on the first photo **and** the evidence, every time · B = OpenCV alone · C = OpenCV then the model (the shipped gate).

Headline — the 50 test-half cases of the model sample (one fixed 54-case sample drawn before any model call; calls, time and money cover all 54 checks; C re-run after the quality-rule change, A and A′ not re-run because the model and its inputs did not change):

	A	A′	C before the fix	C after the fix
bad photos refused (of 33)	11	19	31	**30**
good photos wrongly refused (of 17)	3	3	8	**4**
… without the 2 mislabelled same-place pairs (of 15)	2	1	6	**2**
model calls for 54 checks	54	54	24	31 (14 two-photo)
time p50 / p95	2 989 / 3 738 ms	3 689 / 5 027 ms	95 / 5 159 ms	2 920 / 4 556 ms
$ per 100 checks (token count × list price)	1.27	1.80	0.71	0.89

Sources: `eval/results/compare.json` (A, A′, C after: `slices["test half"]`, `perf`) and `eval/results/compare-before-glare-fix.json` (C before). The "without" row excludes OnQ 04→05 and Commons c09→c10, whose second photo cannot show the first photo's step (label audit in `docs/EVAL.md`); counted from the `rows` of `llm-sample.json`, `llm2-sample.json`, `combined-sample-before-glare-fix.json` and `combined-sample.json`.

- **Where C's lead is, and where it is not.** On real look-alikes of another place, A′ — the model shown both photos — catches them as well as C (6 of 6 in the sample); in C they reach the same two-photo check. C's edge over A′ is what a model cannot see in one or two photos: an earlier photo sent again, nothing changed in the work zone, a change in a do-not-touch zone, overexposure and a work zone cut off — C refused all of those in the sample without a model call, A′ caught 0 of 2 no-change, 0 of 2 do-not-touch, 3 of 6 replays and 1 of 4 overexposed or cut-off cases.
- **C against A:** they decided differently in 20 cases and C was right in 19 (`compare.json` → `c_differs_from_a[].C_right`); the one wrong case is the mislabelled c09→c10. A′ agreed with C in 9 of the 20 (`A_prime_agrees_with_C`). Each case with its pair of photos and both answers: `eval/results/compare.md`.
- **Who refused C's bad photos** (whole sample, 34): OpenCV for the reason the case was built for 22, the model 11, OpenCV for another reason 1 (`compare.json` → `c_caught_by`).
- **The price of fewer false refusals.** After the fix more photos reach the model: 31 calls instead of 24, a median of 2.9 s instead of 0.1 s, $0.89 instead of $0.71 per 100 checks. Per call: one photo $0.0129, two photos $0.0181 (`compare.json` → `per_mode`).
- **Still open — bad photos that pass C:** the mirror-image look-alikes (2 in the sample) pass every configuration, A, A′ and C; the real resent photo with a circle drawn on it (c22→c23) passes, because OpenCV counts the circle as a change and the model accepts it; the smeared trap frame from our own video (f0195) passes OpenCV's blur check.

OpenCV alone on all 548 cases (`eval/results/opencv.json`, `summary.by_split`, after the fix): test half — 130 bad photos blocked, 6 passed and 38 sent to the two-photo check; 2 of 66 good photos wrongly blocked; p50 / p95 68 / 111 ms. Calibration half — 141 blocked, 7 passed, 55 sent on; 1 of 103 good photos wrongly blocked. **Unmodified real photos:** 0 of 56 refused (`eval/results/originals.json`: 6 OnQ, 40 Commons, 10 good video frames; before the fix 8 of 56 were refused, mostly as glare). Of 4 video frames blurred by eye, OpenCV refuses 1 (the whip pan); the other 3 go on to the model.

How to read these numbers. The model figures rest on 2 cases per synthetic class and 1–6 per real class, one run each; the model is not deterministic, so read them as examples, not rates. Synthetic bad photos are built to match the checks, so OpenCV's high catch rate on them is expected by construction; the informative numbers

are the real classes, the false refusals and the unmodified photos. The share of bad photos in the set is constructed, not observed in use. Timings are from a shared Mac.

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7. Limitations and failure cases

- **Real re-shots do not line up.** A photo of the same place from another angle or after cables moved often keeps too few SIFT inliers to confirm the place. Since `793b987` it costs a two-photo model call instead of a refusal; before that, OpenCV refused 3 of 5 such pairs in our set (`eval/results/opencv.md` at commit `ff97511`). OpenCV alone can no longer reject a photo of another room; only the model can, through `same_place`.
- **Parallax.** On the real 01↔02 pair, alignment left 37 % of the frame "changed"; such maps are marked `unreliable` and decide nothing, so `no_change` and `touched_avoid` fire only for photos taken from near the first viewpoint (`docs/PERCEPTION.md`).
- **Zones come from a model.** Work and do-not-touch zones are the Marker model's boxes, which are rough: they drift, run oversized and can overlap (`src/stepspotter/marker.py`, `data/demo/agentic-opencv/NOTES.md`); a wrong zone makes a right change look wrong.
- **Thresholds were set on six photos of one house** and are not a blind test; the scene split shows whether results differ by scene, not field accuracy.
- **Faces:** checked on one public-domain portrait (`tests/test_privacy_faces.py`); no detection rate is claimed.
- **One instance, local disk.** Jobs live on the App Runner instance and are lost on every redeploy, pause or instance replacement (`docs/OPERATIONS.md`).
- **Judging capacity.** The global daily caps (6 jobs, 30 photo checks and 60 agent turns a day, shared by everyone) protect the budget and can run out on a busy judging day; the offline test suite and the eval harness need no AWS account.
- **Misses that remain** (`eval/results/opencv-failures.json`, pictures in `eval/results/figures/opencv-failures-01..02.jpg`): mirror-image look-alikes pass everything; a resent photo with a drawn circle passes; the smeared trap frame f0195 passes the blur check; since the glare fix 11 of 23 synthetic glare discs pass OpenCV and are left to the model; 3 good synthetic photos are still refused (`touched_avoid` twice, `blurry` once).
- **The step card is always drawn on the job's first photo.** A step that happens somewhere else (step 2 of the recorded run is at the office jack) shows its boxes on the first photo.
- **Recorded run pictures.** The change-heatmap captions in the pictures of the recorded agent run were drawn before the caption fix (`3f75518`), which wraps long captions instead of cutting them.
- **Model numbers are small.** 2 cases per synthetic class, one run each (§6).

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8. Responsible use

- **Scope.** Low-voltage, low-consequence jobs only. The planner refuses mains wiring, gas, roofing and structural work with an empty step list (`planner.py:26`). That refusal is a model judgement, not a certified hazard classifier.
- **The person stays in control of risk.** A change in a do-not-touch zone stops the run for a person's approval (`interrupt`, not a cancel), and a hazard seen by the model stops the run too (`gate.py:76`). A refusal always comes with the reason and one instruction; the stuck rule keeps a false positive from trapping anyone.
- **Fail closed.** A photo whose place OpenCV cannot confirm closes a step only if the model says it is the same place (`guide.py:449`, `gate.py:153`).

- **Privacy.** Faces are blurred in the stored file before any model sees it; each report says whether that check ran (`faces_checked`). Photos live only on the instance's ephemeral disk, are not copied to S3 or a database, and are gone on every redeploy. Model-invocation logging is not turned on. CloudWatch gets perception metrics and decisions, no image data; rate-limit rows carry a salted hash of the caller address (`docs/OPERATIONS.md`, `src/stepspotter/store.py`).
- **Spend and abuse.** Per-address and global daily caps, a kill switch (`STEPSPOTTER_PAUSED=1`), a Budget alert and a request-flood alarm.
- **Least privilege.** The instance role can call one model and write one metric namespace; the build role can push to one repository (`infra/template.yaml:271`, `:393`).
- **Data rights.** Our own photos are published with the owner's permission; the face test photo is public domain (`tests/fixtures/ATTRIBUTION.md`); public photos in the evaluation carry their licence and attribution (`eval/cases/public/ATTRIBUTION.md`). The repository's MIT licence does not cover third-party photos.
- **Honest results.** Failures are published next to successes, and every number traces to a committed results file.

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9. Disclosure

StepSpotter was built on 9–14 September 2026 for the AWS "Agents for Humans" hackathon on Devpost, where it won the Everyday Agents Bronze award (announced 14 October 2026). It was also submitted to the AI Builders hackathon, whose judging has not finished. The original code is public at https://github.com/retailbox-automation/stepspotter, commit `80979af` (pushed 2026-09-14). It is imported here as the first commit, `295e3f3`, with the text copied from manufacturers' manuals removed (the bundled manual cache keeps only links and page numbers).

Everything OpenCV 5 — the perception package, its integration into the gate and the verifier, the face blurring, the evaluation harness, the AWS stack, the diagrams and this report — was built on 21–24 October 2026, inside this competition's window: commits `9cc6ca1` through **{{FINAL:head commit}}** (`git log --reverse 295e3f3..HEAD`).

We read the Official Rules for a clause on previously submitted projects and found none. The nearest is the timeline: «From August 26 through October 26, teams will build, evaluate, document, and submit their final projects.» The rules also say «Disclosed agents and automated programs that are integral to the submitted project are allowed and encouraged».

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10. How AI was used

- **In the product:** Claude Sonnet 4.6 on Amazon Bedrock plans the steps, draws the zones and checks the photos OpenCV lets through (one photo, or two photos with `same_place`); in agent mode the Strands Guide agent decides which tool to call next; a tool-less agent answers questions typed into the chat. OpenCV runs before the model on every evidence photo, and the gate sits under every tool call.
- **In development:** the code, tests, infrastructure and these documents were written with AI coding assistants, directed and reviewed by the author. The author chose the product and its rules, took the photos, decided what to publish and approved every spend and every outward step.

Details: `docs/AI-USE.md`.

Reproduce

```
git clone https://github.com/retailbox-automation/stepspotter-opencv && cd stepspotter-opencv
python3.12 -m venv .venv && .venv/bin/pip install -e ".[dev]"
tools/fetch_models.sh                                  # YuNet face detector, SHA256-checked
.venv/bin/python -m pytest -q -m "not integration"     # offline: 433 passed, 1 skipped
.venv/bin/python eval/build_set.py && .venv/bin/python eval/run.py --config opencv
infra/deploy.sh --yes                                  # needs an AWS account; see docs/DEPLOY.md
```